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Palaeomagnetic investigation of Middle Devonian limestones of Algeria and the Gondwana reconstruction

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SUMMARY

We have carried out a palaeomagnetic investigation of the upper middle Devonian marly limestone of the Hazzel Matti Formation outcropping at Meredoua in the northern Ahaggar in the Sahara Desert. The Givetian age of the formation is well constrained by palaeontological arguments. The formation has been deformed later than the Moscovian (upper middle Carboniferous). In the altered yellowish samples and in the heterogeneous samples made of mixed fresh and altered zones, several obviously secondary magnetization components are present. In the freshest blue-grey samples a SE down-dipping component can be isolated, particularly when a combined thermal and AF treatment is applied. This component, observed in 53 samples from three sites, passes the fold test of McFadden & Lowes (1981) and McFadden & Jones (1981) at the 95 per cent probability level. The mean direction for the three sites after tilt correction is: $D_S = 118.9°$, $I_s = 34.2°$, $k = 378$, $\alpha_{95} = 6.3°$, which corresponds to a south pole (MER) situated at $\phi_p = 61.7°$, $\lambda_p = -16.2°$, $A_{95} = 4.2°$ and places the site area under a palaeolatitude of $-18.8°$. However, the best grouping of the directions is achieved after 85 per cent of unfolding using a statistical method based on a bivariate extension of the Fisher's statistics (Legoff 1990; Legoff, Henry & Daly 1992). Such a tilt correction changes the pole position by 1.5°. As the α_{95} and K curves obtained from stepwise unfolding display a broad minimum and maximum value respectively, and because there was probably slight original syn-sedimentary dips, it is likely that 85 per cent of tilt correction is not significantly different from complete unfolding.

The magnetic behaviour of the samples upon thermal and AF demagnetization, the evolution of the initial susceptibility upon thermal treatment, hysteresis cycles and investigation of the magnetic mineralogy through microscope examination and electron microprobe analyses lead to the following conclusion: greigite was probably a primary magnetic mineral formed during the diagenesis of the rock. However, its primary magnetization has been totally replaced by a VRM, except perhaps in rare samples in which the SE component can be found in the narrow range of 300–350 °C. Magnetite is thought to carry the SE magnetization component and may have formed early during the diagenesis or soon after greigite.

To consider the fold test negative would assign the MER pole a post-Moscovian age and place it in a situation inconsistent with the other poles of the West Gondwanian APWP, unless all the middle Carboniferous and probably lower Permian African poles are rejuvenated. We rather favour the interpretation of a positive fold test, which however does not constrain the magnetization to be of primary Givetian age; the SE component may as well have been acquired any time between the Givetian and at least the late Moscovian. Assuming that the MER pole has a Givetian age implies that the Rheic Ocean, which separated Laurussia from Gondwana in the early Palaeozoic, was almost closed in the middle Devonian, in agreement with palaeobiogeographic arguments; but it increases the discrepancy

between the APWP of respectively West and East Gondwana. An intermediate lower Carboniferous age might be in better agreement with both African and Australian data. Clearly, however, more Gondwanian poles are needed to test the reliability of both West and East Gondwanian APWP, and check simultaneously the possibility of relative displacements between these two blocks during the Devono-Carboniferous time period.

Key words: apparent polar wander path (APWP), greigite, magnetite, middle Devonian (Givetian) limestone, palaeomagnetism, West and East Gondwana.

1 INTRODUCTION

The convergence of Gondwana to Laurentia in the course of the Palaeozoic, which leads to the Variscan collision in Permo-Carboniferous times, is the key to the interpretation of the tectonic pattern during the whole Palaeozoic of the presently Atlantic-bordering continents. For this time period, information on relative plate motion can be gained from tectonics, palaeoclimatology, palaeobiogeography and lithofacies study, but only palaeomagnetism is theoretically able to quantify the latitudinal displacements. In practice, however, many causes can distort the palaeomagnetic information, among which the most widespread are the determination of the age of magnetization and the inadequate knowledge of the tectonic setting of the studied area.

This explains why so many palaeomagnetic studies devoted so far to the palaeomagnetism of Gondwana have given so many dispersed poles for a given age, and why there is still no clear trend in the apparent polar wander path (APWP) in the Palaeozoic era. In fact, based on the few reliable palaeomagnetic data for Gondwana (see reviews by Livermore, Smith & Briden 1985; Hargraves, Dawson & Van Houten 1987; Bachtadse & Briden 1989, 1990, 1991), and palaeoclimatological data (see reviews by Caputo & Crowell 1985; Veevers & Powell 1987; Scotese & Barrett 1990), the south palaeomagnetic pole for Gondwana would have settled somewhere close to the north or north-west coast of Africa in the Ordovician times, and east or south-east of South Africa in the Permian times. The APWP remains in doubt from Ordovician to Permian times, between the two optional paths of Morel & Irving (1978). Arguments for path X, connecting more or less directly through Africa the lower Palaeozoic poles to the upper ones, have been put forward by Hurley & Van der Voo (1987) on upper Devonian rocks from western Australia; path Y, which adds a looping hairpin south-west of South Africa in the Silurian times has been supported by the Silurian pole of Hargraves *et al.* (1987) on ring complexes of Niger and by two Devonian poles from south-eastern Australia (Schmidt *et al.* 1986; Schmidt, Embleton & Palmer 1987).

The Siluro-Devonian interval is therefore a critical period of time for which reliable and conclusive palaeomagnetic poles are lacking, particularly for Antarctia and South America, but also for Africa. For these reasons, we have undertaken a palaeomagnetic study of middle Devonian rocks from the Sahara Desert, North Africa.

2 GEOLOGICAL SETTING

The present work deals with a marine marly limestone, the Hazzel Matti Formation of Meredoua (Legrand 1967), deposited in the cratonic Palaeozoic basin bordering the north of Ahaggar, in the Sahara Desert of Algeria (Fig. 1), in the middle Devonian times.

At the end of the pan-African orogeny, the Precambrian shield of Ahaggar was crossed by deep submeridian transcurrent faults (Caby 1968) which fragmented the northern and southern Ahaggar basins into N–S elongated smaller ones. Since then, these faults have been reactivated many times, particularly in the lower Palaeozoic (Beuf *et al.* 1971), when they produced vertical movements generating horsts and grabens. These faults affect the basement and the lower Palaeozoic cover. They die away in the middle Palaeozoic formations, inducing slight tilting of the strata perpendicular to the fault strike, in the near vicinity of these faults (Beuf *et al.* 1971). In the northern Ahaggar basin, the filling up sequences of the marine middle Devonian display a NW Direction of lateral progradation (Bertrand-Sarfati, Fabre & Moussine-Pouchkine 1977).

The upper middle Givetian age of the Hazzel Matti Formation of Meredoua is precisely known by the widespread occurrence of *Maenioceras terebratum* faunas (Moussine-Pouchkine in Fabre 1976; Bertrand-Sarfati *et al.*

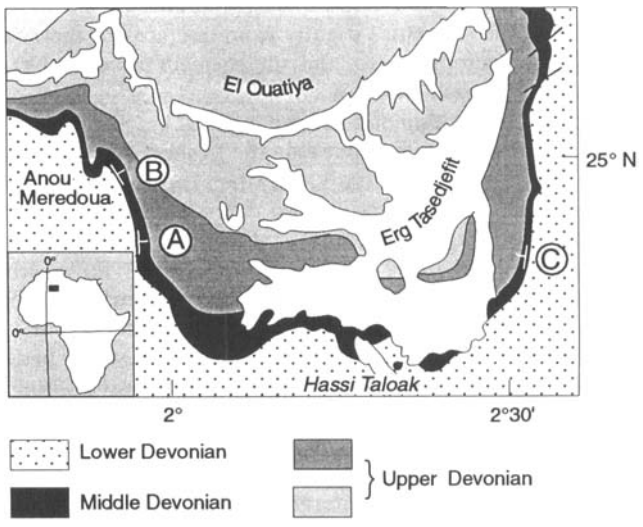


Figure 1. Geological map with the sampling sites.

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1977). The Formation reaches a total thickness of 15 m and is made of a sequence of fissured thin beds about 10–15 cm thick. The rock has a yellow mustard to reddish-yellow colour at the outcrop, but is grey to blue-grey in its freshest parts.

Sedimentological and microscopic observations indicate that the rock was initially deposited in a quiet subtidal reducing environment which was sometimes interrupted by time intervals of more oxidizing conditions. Bioturbation was probably important and responsible for the macroscopic heterogeneity observed in the rock between zones homogenized by the bioturbation on the one hand, and more heterogeneous detrital ones, free of bioturbation, on the other. The presence of thin unbroken shells is indicative of a limited compaction during the diagenesis.

The series have been gently tilted in a wide synclinal structure in the upper Carboniferous or early Permian, as a consequence of the Variscan orogeny developed in northern Algeria and in Morocco. The tectonic tilting is post-Moscovian in age (Donzeau, Fabre & Moussine-Pouckine 1981), as the whole sedimentary sequence up to the Moscovian is deformed without structural unconformity and covered further north by horizontal Cretaceous sediments. Another minor tectonic phase of late Jurassic age (Conrad 1981) has also been identified in some places of the northern Ahaggar and probably accounts for limited reactivation of previous faults or possibly for part of the observed tilting.

3 SAMPLING AND EXPERIMENTAL PROCEDURE

Seventy-three oriented minicores of standard 1 in. (=2.54 cm) diameter and variable length (from 4 to 10 cm) were drilled at three sites of different dips on both sides of the syncline (Fig. 1). The geographic coordinates and the interval of dip variation at each site are:

Site A: coordinates: 24.93°N; 1.93°E;
dip azimuth: 70–85° dip angle: 8–12°
Site B: coordinates: 24.95°N; 1.92°E;
dip azimuth: 86–109° dip angle: 25–36°
Site C: coordinates: 24.90°N; 2.50°E;
dip azimuth: 270–290° dip angle: 5–10°.

In detail the dip varies slightly from one area to the next at a given site; therefore, the tilt correction was applied individually to each group of samples.

One to four specimens of standard 2.2 cm height were sliced out of each sample, but only the freshest one was used in the palaeo-directional study, or, when two specimens of a sample were demagnetized by two different techniques (AF and thermal), only one mean direction was kept for calculations. The other specimens were saved for microscopic and rock-magnetic investigations, provided they looked similar to the previously selected one; this was often not possible due to the fissuring of the limestone beds; alteration is often pervasive and a single core generally shows a more or less continuous gradation from fresh to highly altered.

Site A provided the freshest and the most homogeneous blue-grey samples. The alteration is slightly more pronounced at Site B where the samples are often dull grey and heterogeneous. At Site C, the samples are never totally fresh but rather exhibit an heterogeneous mixture of well-delimited moderately fresh grey and yellow to reddish-brown zones.

All the samples were subjected to a 15 day viscosity test, as described in Prévot (1981). The remanent magnetization was measured on a CTF cryogenic magnetometer. For viscous samples, the intensity changes of the components occurring during a measurement were visualized on a screen after each demagnetization step. This procedure allows the operator to wait until the component stabilizes and it proved to be very useful in the elimination of the VRM fraction acquired during handling in the laboratory.

Stepwise thermal demagnetization was performed on all the samples in a furnace whose residual field is less than 20 nT. AF demagnetization was also performed up to 200 mT on specimens in a laboratory-built AF demagnetizer described in Legoff (1985) and equipped with an automatic self-compensating ARM system. In most cases, however, a combined thermal and AF treatment was the most efficient technique to isolate the magnetization components (Daly 1981).

The susceptibility was measured with a Bartington MS2 magnetic susceptibilimeter after each temperature step, to monitor chemical alteration of the magnetic minerals upon heating.

The directions of the magnetization components were computed by principal component analysis (Kirschvink 1980) and represented on orthogonal diagrams. The residual and the difference vectors were plotted on equal-area projections. The Fisher's statistic (Fisher 1953) was used to calculate the mean directions and the associated pole positions.

Hysteresis cycles were run on 18 minicores of both fresh and altered samples ($\phi = 1.6$ cm; h = 1.7 cm) at the Geomagnetism Laboratory of Saint-Maur-des-Fossés. A small translation inductometer settled between the pole pieces of an electromagnet allows the induced and remanent magnetization to be measured up to a maximum inducing field of 1.6 T. The remanent magnetization is measured after compensation for the electromagnet residual field.

4 RESULTS

(1) *NRM intensity*: The intensity of magnetization, J_r, covers more than two orders of magnitude at the three sites. It varies from 1.2×10^{-7} to 1.3×10^{-6} $Am^2 kg^{-1}$ at Sites A and B, and from 1.5×10^{-6} to 2.7×10^{-5} $Am^2 kg^{-1}$ at Site C, exceeding 10^{-4} in few samples. Clearly the more altered the samples are, the stronger their magnetization intensity.

(2) *Initial magnetic susceptibility*: Before heating, the initial susceptibility, K, follows the same trend but with a narrower variation range; it varies from 2.4×10^{-8} to 9.6×10^{-8} $m^3 kg^{-1}$ at Sites A and B, and from 6.8×10^{-8} to 1.8×10^{-7} at Site C.

(3) *Viscosity test*: The VRM acquired in the 15 day experiment is always negligible for the Site C samples; here the viscosity coefficient, v, varies between 0 and 11 per cent; it is generally high in the samples from Sites A and B ($5 \leq v \leq 85$ per cent) where, in a few cases, it reaches the 'stable NRM' intensity ($v = 100$ per cent), that is the intensity of the original NRM from which the 15 day VRM is substracted.

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4.1 The magnetization components

The samples behave differently according to their alteration degree, and the heterogeneous ones, made of fixed fresh grey and altered yellow areas, include more components than the homogeneous ones. The spectra of unblocking temperatures, T_{ub}, and unblocking fields, H_{ub}, shown by the demagnetization diagrams, together with the susceptibility changes during heating, provide useful information on the nature of the magnetic carriers.

The freshest grey-blue samples of Sites A and B generally display only two components of magnetization (Figs 2 a and c): the first one has low unblocking temperatures (T_{ub} usually <250 °C) and a direction close to the present axial dipole field one (PDF component, Fig. 4). Because it is also found in the more viscous samples, we interpret it as a field VRM. The secondary nature of this component is evidenced by the better grouping of the directions in the geographic than in stratigraphic coordinates. Although it is generally intense relative to the NRM (Fig. 2a), its direction has no interest here and thus will not be considered any further.

The initial susceptibility, K, remains constant up to

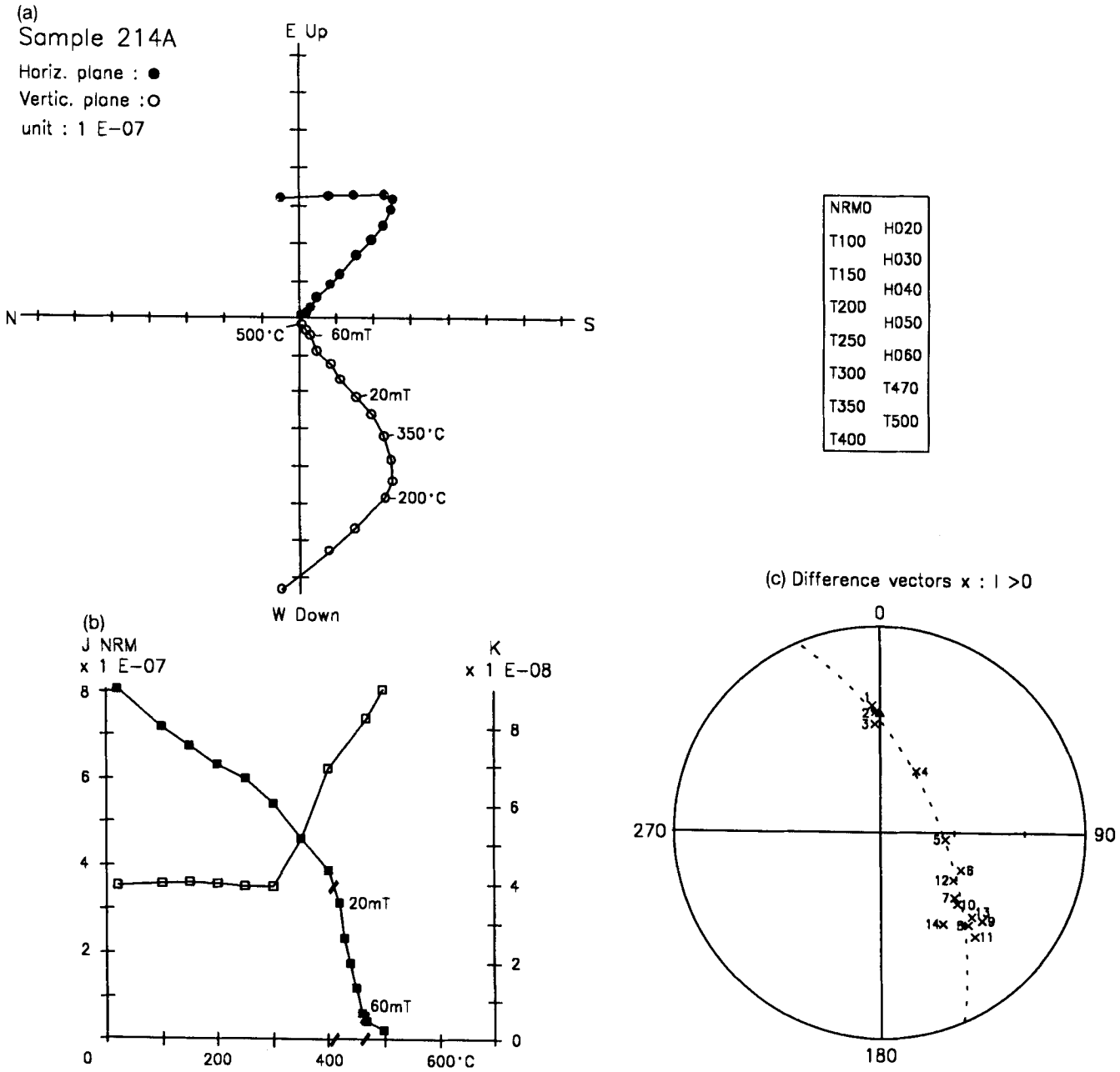


Figure 2. Typical demagnetization diagrams of a sample from Site A displaying two magnetization components, a VRM and the SE component. (a) Orthogonal projections in geographic coordinates; (b) changes of the remanent magnetization intensity, J_{NRM}, and of the initial susceptibility, K, upon heating treatment; (c) evolution of the difference vectors in a lower hemisphere equal-area projection. The insert displays the successive demagnetization steps; T and H mean respectively thermal and AF treatment.

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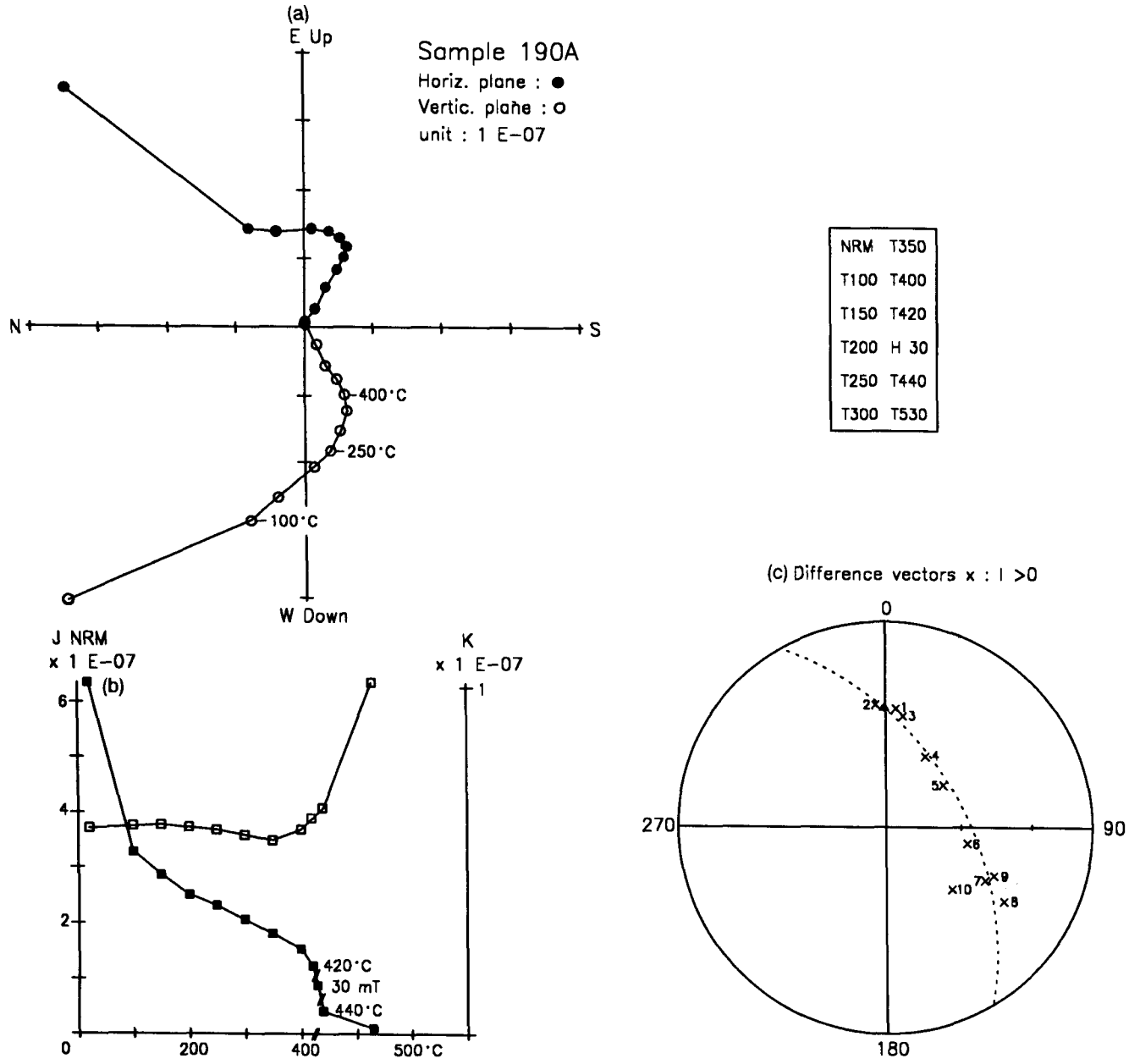


Figure 3. Same legend as in Fig. 2 for a sample from Site B containing a mixture of fresh and altered zones: a third magnetization component with very low unblocking temperatures is present; here, thermal, then alternating field and finally thermal treatment are successively used.

300–350 °C (Figs 2b and 3b), then starts to increase with the temperature, indicating a change in the magnetic mineralogy; this change results either from the formation of a magnetic mineral from initially non-magnetic ones, or from the inversion of an original magnetic mineral into a more strongly magnetic one. Ferrimagnetic pyrrhotite is known to invert into magnetite at around 350 °C (Dekkers 1989a; Rochette *et al.* 1990) or 400°C (Bina *et al.* 1991). The spontaneous magnetization of Magnetite is five to six times higher than that of monoclinic pyrrhotite, which may be a possible explanation for the susceptibility increase above 350–400 °C.

To avoid the production of an eventual secondary TRM acquired above 350 °C by the newly crystallized magnetic minerals during cooling in the residual field of the oven, the samples were subsequently AF demagnetized up to the highest possible level (50–70 mT) above which the directions became too scattered to be interpretable, although 10–20 per cent of the initial NRM still remained. At this stage, the thermal treatment was resumed until 500–570 °C; above this temperature, remaining directions become erratic. The susceptibility increases sharply: it doubles around 500–550 °C, but the direction remains constant and converges toward the origin, from the beginning of the AF treatment up to the highest temperature steps (Figs 2a and 3a). Therefore, the south-east and down-dipping direction of this last magnetization component, SE (Figs 2c and 3c), cannot be suspected to be an artefact due to a spurious TRM

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acquired during the thermal treatment by the magnetic phase crystallized above 350–400 °C. The SE magnetization component is represented by a single polarity.

The combined demagnetization technique that uses thermal, AF and thermal treatment again, proved to be the more efficient one to isolate the SE component. However, this procedure leads to demagnetizing curves, $J_r = f(T, H)$, whose interpretation is not straightforward because of the scale discontinuity along the abscissa between thermal and AF and vice versa. Here, the magnetization drop associated with the AF treatment in Figs 2(b) and 3(b) would only be a smooth decrease if the scale of the AF intensity in abscissa were expanded.

When only thermal demagnetization is used, about 10–30 per cent of the initial NRM remains after 500 or 520 °C, but then a precise SE direction cannot be determined because the residual vectors scatter, possibly due to the addition of a minor parasitic TRM acquired by the newly formed magnetic phase.

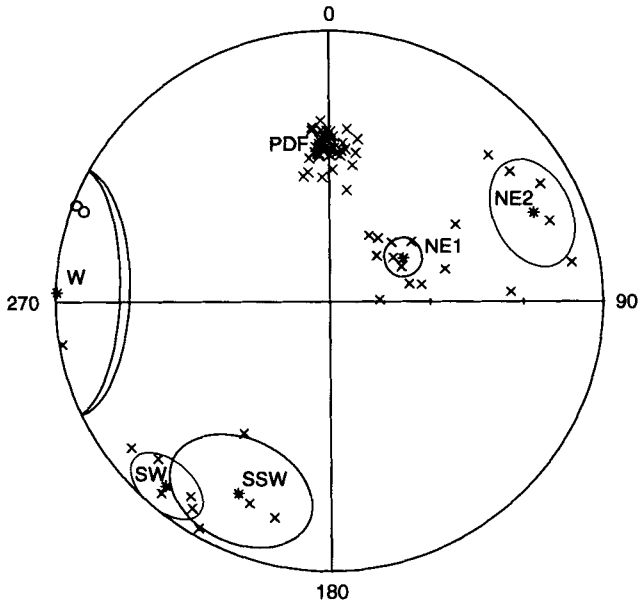


Figure 4. Lower hemisphere equal-area projection of the direction, in geographic coordinates, of the various secondary magnetization components found in the Meredoua marly limestones and giving a negative fold test (see also Table 1). The inclination of the dipole field (±42.9°) falls in the middle of the PDF directions.

Thermal and AF demagnetization indicate that the SE component is characterized by unblocking fields, H_{ub}, spanning the interval of 10–70 mT and unblocking temperatures of between 250 and 550 °C. This is consistent with fine-grained magnetite carrying the SE magnetization.

We note that in the rare totally altered samples of Sites A and B, the SE component is not found.

In the few samples from Sites A and B composed of a mixture of fresh grey and altered yellow zones, a third component of magnetization is observed (Figs 3a and b). Its intensity relative to the initial NRM is large when the altered zone represents an important fraction of the sample. Its direction, variable from sample to sample, gives no coherent pattern whatever the reference frame. This component is characterized by very low T_{ub}(<120 °C) and very high H_{ub} (>200 mT); this is typical of goethite (O'Reilly 1984; Dekkers 1989b and c), as is the slight fluctuations of the susceptibility upon heating: K decreases between about 250 and 360 °C during the dehydration of goethite, then increases slightly when goethite inverts to Haematite (Dekkers 1988a).

In the heterogeneous samples, found mainly at Site C, the interpretation of the demagnetizations is generally not straightforward, first because the samples usually include from one to three components, and then because their explanation is often problematic. Apart from a widespread but usually relatively small component close to the Earth's present magnetic field direction (PDF, Fig. 4), six other components can be determined in the heterogeneous samples. One is the SE component, and the other five, found in a small number of samples, are represented in Fig. 4 and Table 1. They occur in several possible associations with each other without a general coherent trend; an example of such association is given in Fig. 5(c). Several points can be noticed:

(1) The PDF component is often the last one to be demagnetized (Fig. 5). Its high unblocking temperature spectrum (T_{ub} up to 670 °C) and high unblocking fields

Table 1. The secondary magnetization components which fail the fold test, at the three sites of the Meredoua limestones

name	n	I	D	k	α_{95}	λ_p	ϕ_p	K	A_{95}
PDF	44	42.7	359.3	197	1.5	89.3	258.0	205	1.4
W	4	-1.6	272.3	12	27.1	1.7	270.8	13	26.8
SW	5	10.0	221.3	64	9.6	-39.5	303.9	74	9.0
SSW	4	23.4	205.3	26	18.4	-44.9	326.1	44	14.0
NE1	12	64.6	58.3	60	5.7	39.4	49.7	30	8.0
NE2	6	20.7	65.9	25	13.7	26.1	92.2	26	13.3

Name: component's name; n: number of samples; I and D are respectively the inclination and the declination of the magnetization direction in geographic coordinates, except for the W component which is given in stratigraphic coordinates; λ_p and ϕ_p are respectively the latitude and the longitude of the VGPs; k (K) and α_{95} (A_{95}) are the precision parameter and the 95 per cent confidence angle (Fisher 1953) on respectively the directions and the poles.

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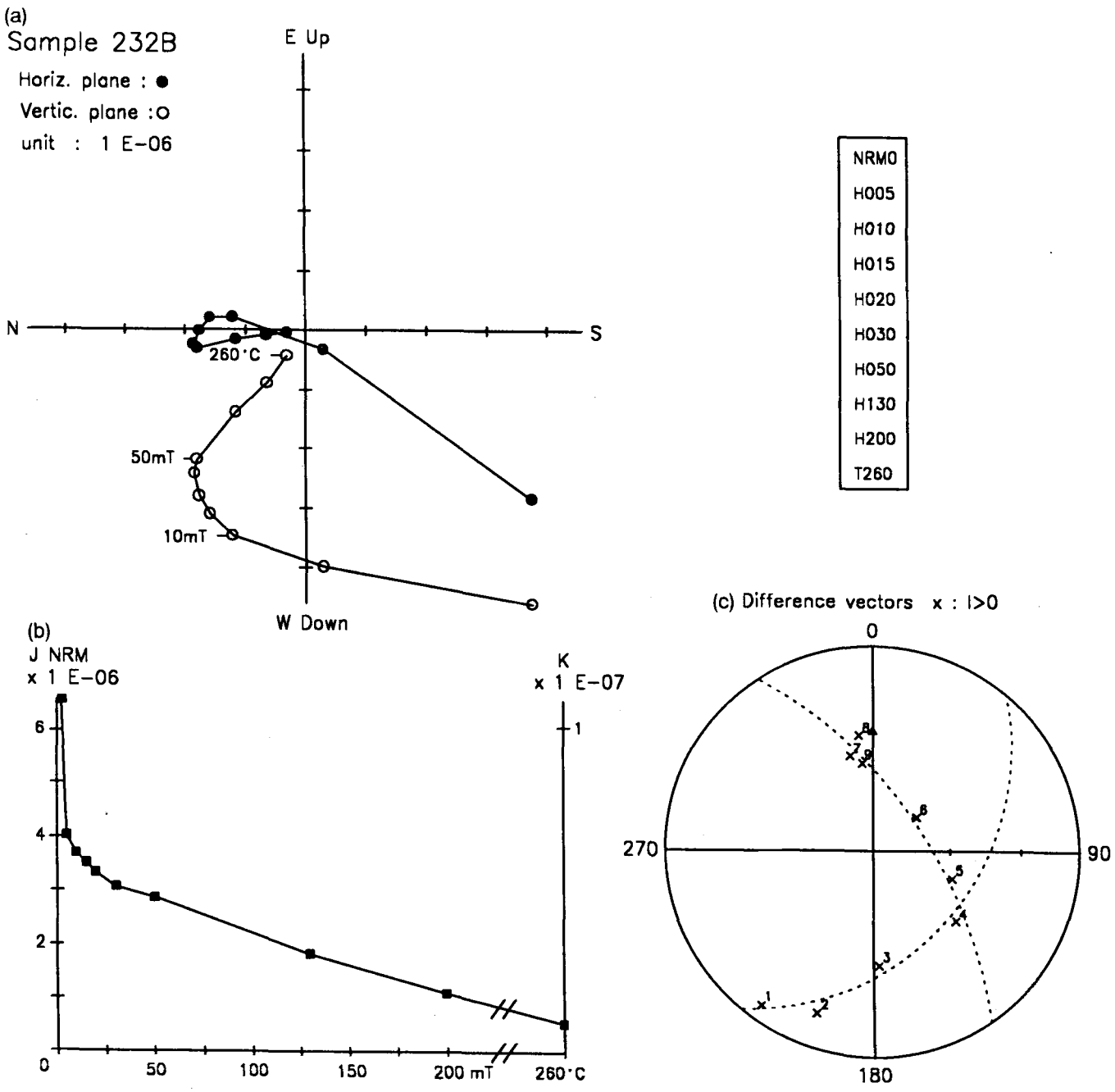


Figure 5. Same legend as in Fig. 2 for an altered heterogeneous sample from Site C containing the SW, the SE and the PDF components with overlapping of the T_{ub} spectra; here, AF followed by thermal demagnetization is used.

($H_{ub} > 200$ mT) suggest the presence of haematite carrying a recent CRM.

(2) The SE component is only found in the heterogeneous samples which include grey nodules well preserved among an altered yellow to reddish matrix. However, this component is not always directly resolved because of the overlapping of its T_{ub} spectrum with those of the other components, but can be determined through the intersection of great circles passing through the difference vectors between successive demagnetizing steps (Fig. 5c), resulting from a mixed AF and thermal treatment.

(3) In several reproducible cases, we observed in two apparently identical adjacent specimens of a common sample that the thermal treatment reveals only one or two magnetization components (Fig. 6c), whereas an AF followed by a thermal treatment distinguishes three directions (Fig. 5c).

(4) The PDF component, observed in 44 samples (Table 1), is very close to the dipole field direction ($I = \pm 42.9°$). It corresponds to a VRM in the fresh samples and to a CRM in the altered ones, but whatever its nature, it never passes the fold test.

(5) The components SW, SSW, NE1 and NE2 are found in few samples, always among the most altered ones, and again none of them passes the fold test. Their corresponding poles (Fig. 9) do not coincide with any of the African poles

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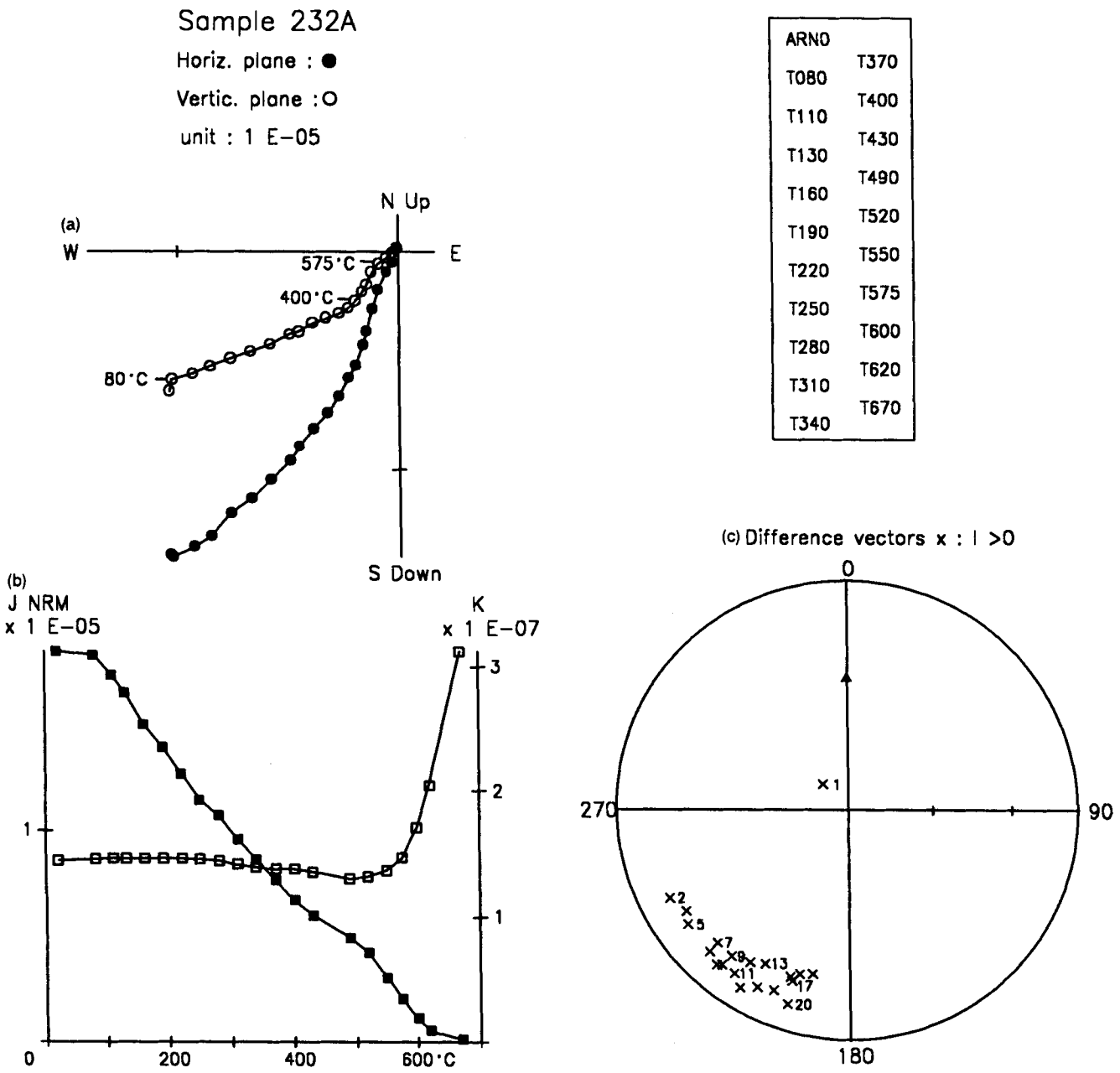


Figure 6. Same sample as in Fig. 5 but on an adjacent specimen displaying two distinct components (SW and SSW); the thermal treatment alone has been used; here, neither the SE nor the PDF can be seen.

younger than Devonian. No obvious explanation can be put forward, but these components are certainly secondary and therefore will not be considered any further.

The SE component could be isolated in 53 samples from the three sites. Its polarity is exclusively reversed. The mean direction calculated for the individual samples is tabulated in Table 2 and illustrated in Figs 7(a) and (b), respectively before and after tilt correction. The mean direction calculated for the three sites is: $D_g = 123.7°$, $I_g = 44.2°$, $k = 33$, $\alpha_{95} = 21.9°$ in geographic coordinates (Fig. 8a, Table 2) and $D_s = 118.9$, $I_s = 34.2°$, $k = 378$, $\alpha_{95} = 6.3°$ in stratigraphic coordinates (Fig. 8b, Table 2). The fold test is positive at the 95 per cent probability level, according to McFadden & Lowes (1981) and McFadden & Jones (1981). The coordinates of the south pole, MER, calculated from the tilt-corrected directions of the three sites are: $\lambda_p = -16.2°$, $\phi_p = 61.7°$, $A_{95} = 4.2°$, $k = 852$ (Fig. 9).

However, a proportional stepwise unfolding of the synclinal using a code developed by M. Legoff and based on a bivariate extension of the Fisher's statistics (Legoff 1990; Legoff *et al.* 1992) provides complementary information (Table 3): from the *in situ* position up to 30 per cent of untilting, the k_y/k_x ratio exceeds the critical value of F $2(N-1)$, $2(N-1) \sim 1.6$ given by the tables of variance ratios for a set of 53 samples, which confirms that the SE magnetization component has not been acquired after the structural deformation. In contrast, k_y/k_x is lower than the

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Table 2. The SE component: directions and corresponding poles

	N	D_g	I_g	k	α_{95}	D_s	I_s	k	α_{95}	λ_s	ϕ_s	A_{95}	K
						For 100 % of unfolding :							
Site A	23	126.9	42.9	68	3.7	120.2	35.5	77	3.5				
Site B	22	130.8	57.8	33	5.5	118.8	29.7	48	4.5				
Site C	8	116.6	31.5	40	8.8	117.7	37.5	41	8.8				
Samples	53	126.2	47.4	27	3.9	119.2	33.4	53	2.7	-16.5	61.7	2.5	63
Sites	3	123.7	44.2	33	21.9	118.9	34.2	378	6.3	-16.2	61.7	4.2	851
						For 85 % of unfolding :							
Samples	53					119.9	35.4	55	2.6	-16.6	60.6	2.5	64
Sites	3					119.3	35.6	1248	3.5	-16.0	60.8	3.2	149

Same legend as in Table 1; the subscript *g* refers to geographic coordinates and *s* to stratigraphic coordinates after total and partial tilt correction.

critical value when the degree of untilting is between 40 and 100 per cent; consequently, the hypothesis of a rotational symmetry of the directions around the mean cannot be rejected at the 95 per cent level of significance and thus argues in favour of a positive fold test or possibly a syn-folding magnetization. But the fluctuations of k_y/k_x between 60 and 100 per cent of tilt correction do not allow an optimal degree of unfolding to be determined. Therefore, we rather consider the other parameters which behave coherently. The best grouping of the directions is achieved between 80 and 90 per cent of unfolding (minimum value of α_{95}, $\alpha_{95}(y)$, $\alpha_{95}(x)$ and $\alpha_{95}(x)/\alpha_{95}(y)$) with a maximum value of K around 85 per cent (Table 3 and Fig. 10). We note that the K curve displays a broad maximum and the α_{95} a wide interval of minimum values (Fig. 10); this can be expected when the dips are small (Watson & Enkin 1993), so that the differences in the remanence direction distribution do not change dramatically upon progressive untilting.

In this context, is 85 per cent of unfolding significantly

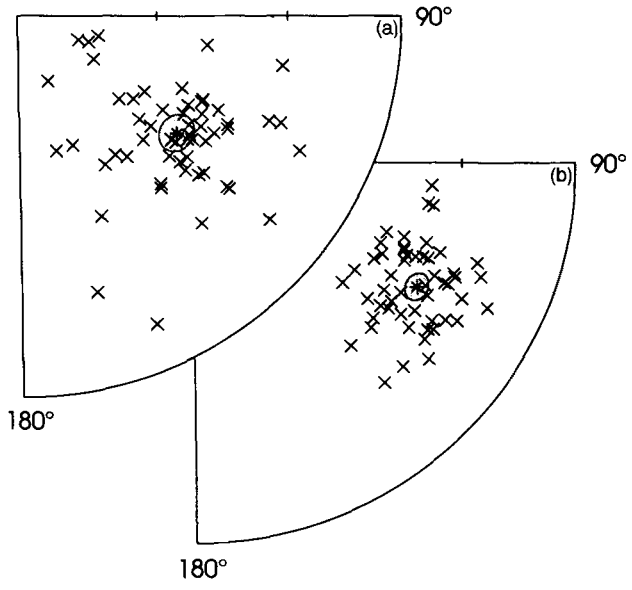


Figure 7. Mean direction of the SE magnetization component for the individual samples (a) before and (b) after tilt correction (see also Table 2).

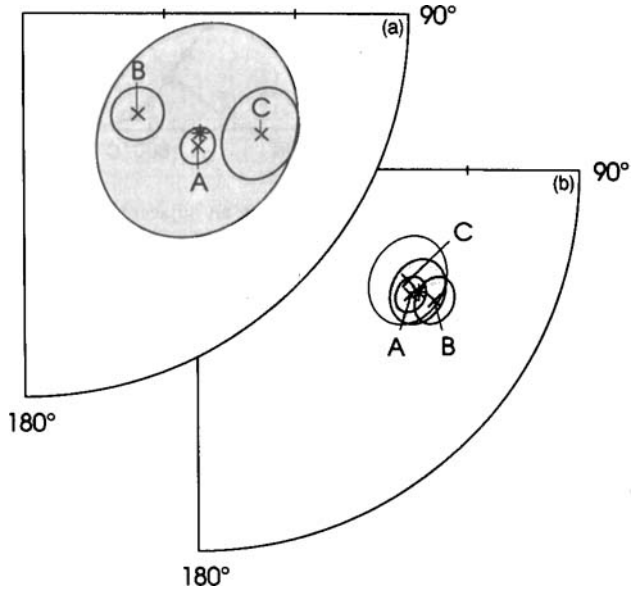


Figure 8. Mean direction of the SE magnetization component for the three sites (a) before and (b) after tilt correction (see also Table 2).

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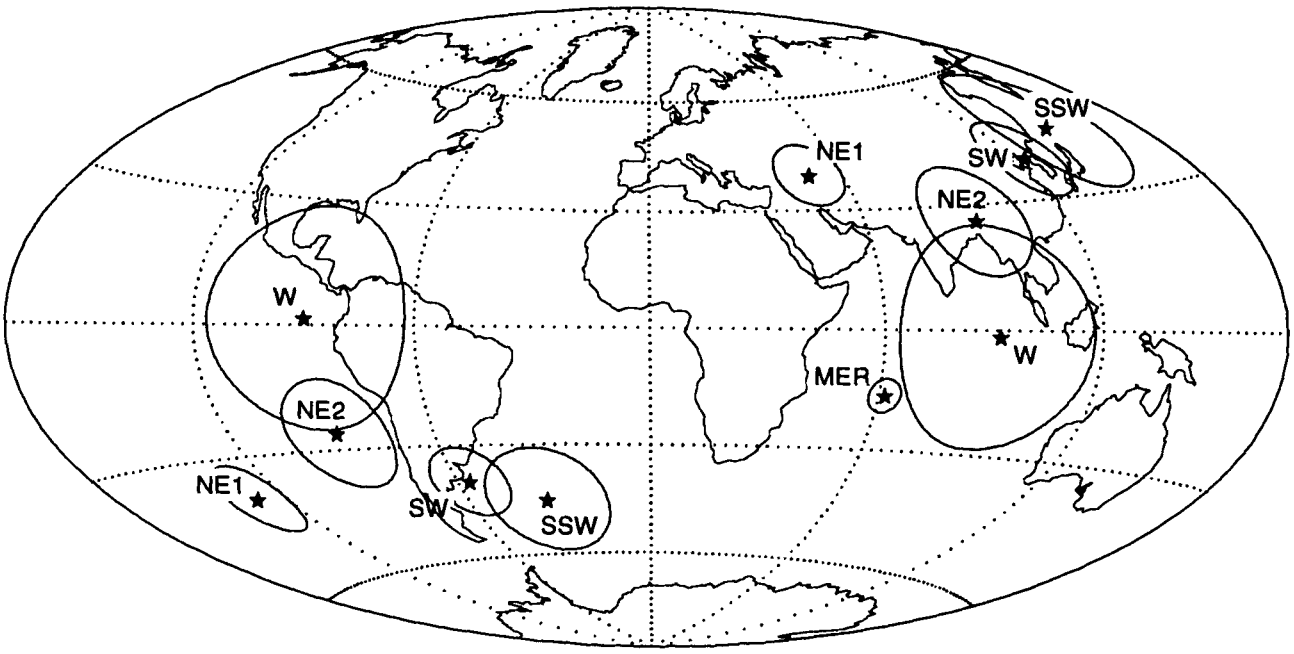


Figure 9. VGPs (north and south) of the various magnetization components found in the Meredoua marly limestones. Only the MER pole is a plausible one for a middle Devonian or more recent age.

Table 3. Bivariate and Fisher's statistic parameters as a function of progressive unfolding for the 53 individual samples displaying the SE component

	BIVARIATE								FISHER			
%	D	I	Omega	K_y	K_x	K_y/K_x	$\alpha_{95}(y)$	$\alpha_{95}(x)$	D	I	K	α_{95}
0	126.2	47.2	167.8	48.7	18.6	2.62	2.8	4.5	126.2	47.3	26.2	3.8
10	125.2	45.9	170.0	49.8	21.9	2.27	2.7	4.1	125.2	46.0	29.7	3.5
20	124.3	44.5	172.7	50.9	26.0	1.96	2.7	3.8	124.3	44.6	33.6	3.3
30	123.5	43.2	176.4	52.0	30.8	1.69	2.7	3.5	123.5	43.2	37.8	3.1
40	122.7	41.8	2.4	53.3	36.3	1.47	2.6	3.2	122.7	41.8	42.3	3.0
50	122.0	40.4	12.5	55.2	41.9	1.32	2.6	3.0	122.0	40.4	46.6	2.8
60	121.3	39.0	27.4	58.2	46.3	1.26	2.5	2.8	121.3	39.0	50.6	2.7
70	120.7	37.6	39.8	61.9	48.7	1.27	2.4	2.8	120.7	37.6	53.6	2.6
80	120.2	36.2	44.9	64.5	49.6	1.30	2.4	2.7	120.2	36.2	55.1	2.6
85	119.9	35.4	44.9	64.8	49.7	1.30	2.4	2.7	119.9	35.4	55.3	2.6
90	119.6	34.7	43.6	64.4	49.5	1.30	2.4	2.7	119.7	34.7	55.1	2.6
100	119.2	33.3	36.0	62.1	48.1	1.29	2.4	2.8	119.2	33.3	53.3	2.6

The magnetization directions at Sites B and C, have been reduced to the Site A coordinates. %: percentage of unfolding. The following parameters are calculated in Legoff (1990) and Legoff *et al.* (1992): Omega: angle between the great axis of the confidence ellipse and the meridian crossing the mean direction; K_x and K_y are the inverse of the variances in the $(x-y)$ plane perpendicular to the mean direction; $\alpha_{95}(x)$ and $\alpha_{95}(y)$ are the axes of the 95 per cent confidence ellipse around the mean direction. K and α_{95} are the usual Fisher's (1953) parameters. Note that the mean direction calculated by means of bivariate and Fisher's statitics are very similar.

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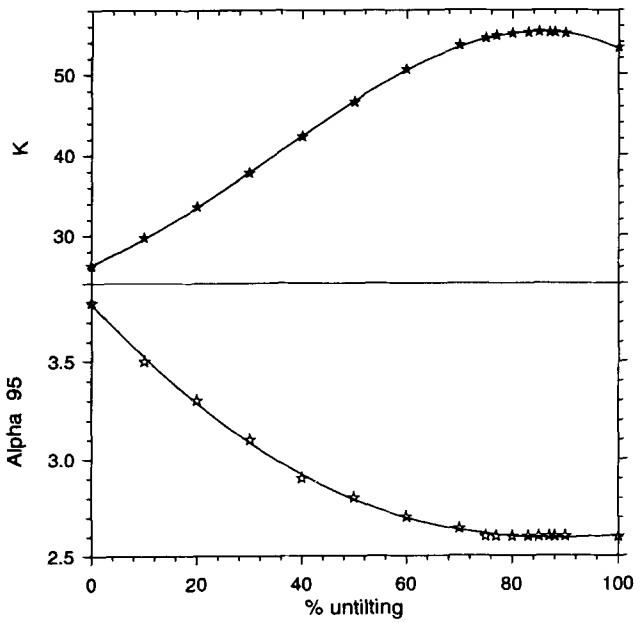


Figure 10. Evolution of the Fisher's (1953) concentration parameter (K) and the half-angle of 95 per cent confidence cone (α_{95}) during stepwise unfolding of the SE magnetization component.

different from total tilt correction? The problem is thus to determine the confidence limits on the optimum degree of unfolding at a given significance level. Watson & Enkin (1993) showed that for structural dips within 5–10°, the 95 per cent confidence interval could span an untilting interval of 60 per cent, which means that 70 per cent of unfolding could not be considered to be significantly different from 100 per cent of tilt correction.

An example comparable to the present study is reported in Chen, .Powell & Balme (1993) on the Brewer Conglomerate in central Australia; here, the best results of the McFadden's (1990) fold test at the 95 per cent significance level are achieved for 96 per cent of untilting, but the optima values of K and α_{95} for their C3 component correspond to 91 per cent of unfolding, whereas these parameters remain approximately constant between 80 or 90 and 100 per cent of tilt correction. Inside this interval, the difference in the mean direction is less than 1°. For the above reasons, the authors consider the 100 per cent corrected direction to be the best estimate of the directions of the primary remanence.

In the present study, the difference in the mean direction between 85 and 100 per cent of tilt correction is 2.2° (Table 3), which corresponds to a change of only 1.5° of the pole position (Table 2). An optimum unfolding of 85 per cent means that the best grouping of the directions is achieved when Sites A and C are restored in a quasi-horizontal position with a dip of 1° while the remaining dip is about 4° at Site B. A residual dip of few degrees may not be significant for the following reasons: errors in the orientation of the samples and on the dip measurements when the number of samples at each site is small (23, 22 and 8 at Sites A, B and C respectively) can also leave a residual dip which in fact did not exist when the magnetization was acquired. Conversely, the hypothesis usually made in palaeomagnetism that sediments are deposited horizontally is not always valid because the sediment deposit is strongly influenced by any pre-existing topography of the basin (see, for example, the syn-sedimentary dips noticed by Hurley & Van der Voo (1987) in the Canning Basin study). As previously mentioned, an original slight dip probably existed at Meredoua, as a result of the combined effects of the NW direction of progradation of the middle Devonian Series (Bertrand-Sarfati *et al.* 1977) and the proximity of the sites (B and C particularly) to N–S syn-sedimentary faults (Beuf *et al.* 1971). For the above reasons, the residual dips obtained after 85 per cent of unfolding are probably not significant so we may accept that the fold test is positive and that the SE magnetization component predates the deformation. But because we cannot exclude totally the possibility that this remanence is syn-deformation, we will subsequently examine both interpretations.

The possibility that the W component could represent the normal polarity of the SE component was examined. However, the W component is isolated in four samples only, having the same dip (no possibility for a fold test), the directions are scattered (Fig. 4) and significantly different ($D = 272.2°$, $I = -1.6°$, $k = 12$, $\alpha_{95} = 27.1°$, in stratigraphic coordinates) from the reversed SE one at the 95 per cent probability level. Furthermore, the W component is found in very altered samples, with no other component and thus is believed to be secondary.

4.2 Magnetic mineralogy

Sixteen polished sections representative of the whole sampling were examined in reflected light. The fresh grey samples are characterized by the occurrence of few but large crystals (10–100 μm) of automorph pyrite, sometimes surrounded by a rim of goethite. This identification is confirmed by the electron microprobe analyses on two fresh samples: the weight concentration recalculated for 100 per cent is Fe = 48.19; S = 51.81; the corresponding empirical formula is $Fe_{1.07}S_2$.

Rare small grains (less than a few micrometres) of another sulfide are also observed, but their size is too small to allow for good quantitative analyses. Only four of these grains, with a maximum of three analyses per grain, could be analysed on the two samples. Although scattered, the weight concentrations are significantly different from that of pyrite; it is Fe = 57.59; S = 42.41, which gives a formula of $Fe_{3.4}S_4$ close to that of greigite (Fe_3S_4).

In the fresh samples, no other opaque mineral could be seen in reflected light, except rare tiny reflective grains ($\ll 1$ μm) too small to be analysed.

In the altered samples, marcassite is sometimes present in the fresh zones and haematite is ubiquitous in the altered ones, either as alteration product of old pyrites, as evidenced by the well-preserved square automorph sections, or as tiny grains spread in the matrix and along cracks. Goethite is often associated with haematite.

4.3 Hysteresis parameters

Hysteresis cycles provide useful quantitative information about the nature and the size of the magnetic carriers, provided they constitute a unique population in the studied

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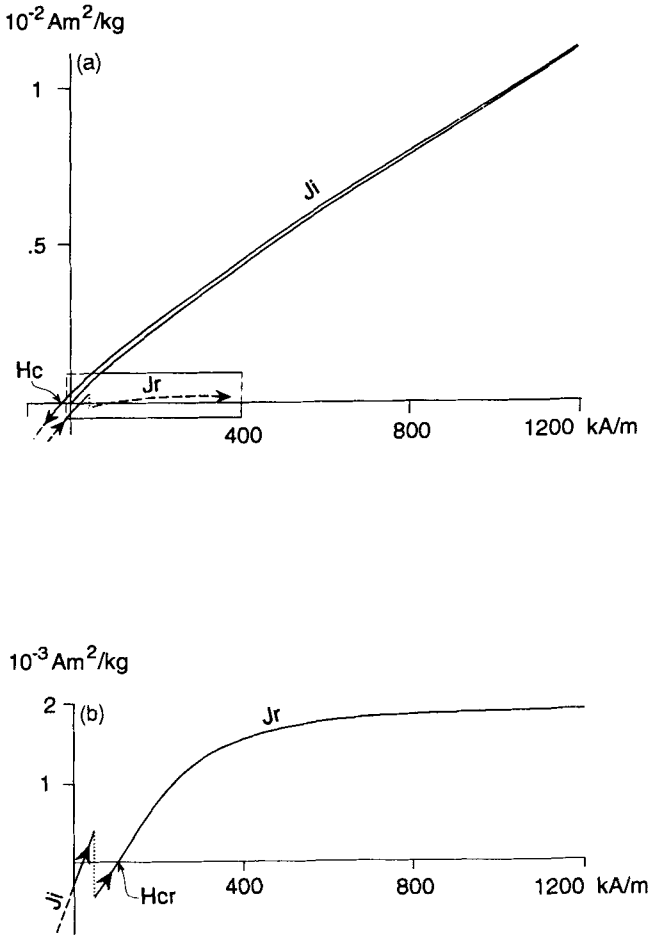


Figure 11. Hysteresis cycle of one of the freshest sample consisting of two magnetization components: the PDF and the SE ones. (a) Acquisition of the induced magnetization, J_i; (b) acquisition of the remanent magnetization, J_r (800 kA m^{-1} = 1 T).

samples. In the present study, where at least two populations of magnetic minerals occur in a single sample, only qualitative informations can be gained from the hysteresis cycles, particularly from the remanence acquisition curve, the induced magnetization of the ferrimagnetic minerals being heavily obscured by the paramagnetism of the matrix, in weakly magnetic sediments.

In the freshest samples (Fig. 11) the saturation is reached between about 0.96×10^3 kAm^{-1} and up to or above the maximum available field of 1.28×10^3 kAm^{-1}. The ferrimagnetic coercive force, H_c, corrected for the paramagnetism varies between 12 kAm^{-1} and 20 kAm^{-1} and the remanent coercive force, H_{cr}, between 52 kAm^{-1} and 78.4 kAm^{-1}. The presence of a highly coercive magnetic phase (hard phase) is thus evidenced by high saturation fields and H_{cr} values, while the existence of a magnetic phase with low coercivity (soft phase) is indicated by low H_c values. This apparent discrepancy between the values of H_c and H_{cr} has been experimentally explained by Day, Fuller & Schmidt (1977) who showed that, in a mixture made of 30–50 per cent of a hard magnetic phase and 70–50 per cent of a soft one, H_c is dominated by and representative of the soft population, while, on the opposite, H_{cr} reflects the hard fraction. The 'soft' population could be magnetite or maghemite, and the 'hard' one, iron sulfide, haematite or goethite.

Hysteresis cycles carried out on a fresh sample previously demagnetized up to 600 °C show that, compared to the unheated fresh sample, the curve of induced magnetization increases faster in low fields, and saturates earlier ($H_{sat} \sim 320$ kAm^{-1}) with J_s and J_{rs} values one order of magnitude higher, while H_{cr} decreased by a factor of two. In agreement with the susceptibility increase observed above 350–400 °C, this demonstrates that a soft highly magnetic fraction has been crystallized at the expense of a hard one. Most probably magnetite has been crystallized from an original Fe-sulfide.

In the altered samples, various behaviours are observed whose interpretation is not easy because of the great number of magnetic components probably implying a greater number of ferrimagnetic phases. Two main conclusions can be drawn from these data: a soft magnetic phase is generally present, as can be seen from the rapid increase of J_i in low fields, suggesting magnetite or/and maghemite; a hard magnetic phase is ubiquitous as expressed by the fact that the saturation is not reached in the maximum field of 1.28×10^3 kAm^{-1} and by H_{cr} values up to 0.48×10^3 kAm^{-1}. Such high values of H_{cr} eliminate the possibility for a ferrimagnetic sulfide to be one of the magnetic carrier (Dekkers 1988b) but rather points to goethite (Dekkers 1989b) or haematite.

5 RELATION BETWEEN THE MAGNETIC CARRIERS AND THE MAGNETIZATION COMPONENTS

A combined interpretation of the information gained from thermal and AF demagnetizations, microscope examination and hysteresis cycles can now be proposed to define the nature of the magnetic minerals related to the magnetization components.

In the fresh samples containing only two components, the first one (VRM) is carried by a magnetic phase which inverts above about 350 °C, which is approximately at the temperature from which the SE component starts to be isolated (Figs 2a and 3a). The second component (SE) is carried by magnetic minerals whose T_{ub} spectra occur between 250 and 550 °C, the two components overlapping between about 250 and 350 °C. The inversion around 350 °C upon heating in air is typical of either ferrimagnetic Fe-sulfide, maghemite or titanomaghemite. The last of these can be discarded simply on the basis of environmental conditions during the sediment genesis.

Maghemite is also unlikely because upon heating in air it inverts to haematite; since the spontaneous magnetization of haematite is about 200 times lower than that of maghemites, the susceptibility should decrease above 350–400 °C instead of increasing, as presently observed (Figs 2b and 3b). Also, the high saturation fields and high H_{cr} values reveal the existence of a hard magnetic fraction in addition to a soft one. If the soft one, carrying the SE magnetization, is magnetite, as strongly suggested by the T_{ub} spectra and supported by the H_{ub} spectra, the hard fraction could be either goethite, haematite or a Fe-sulfide, but not magnetite. Goethite and haematite can be both discounted because their unblocking temperatures are too low (T_c of

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goethite does not exceed 120–130 °C; O'Reilly 1984) or too high (T_c of haematite = 675 °C), compared to what is observed in Figs 2(a) and 3(a). Consequently, the mineral which inverts around 350 °C can only be a ferrimagnetic Fe-sulfide; this is consistent with the presence of pyrite observed and analysed in the fresh samples, and with the existence of greigite suggested by few microprobe analyses. Greigite is common in marine (Morse *et al.* 1987) and freshwater sediments (Snowball 1991) and can be precipitated by bacteria (Mann *et al.* 1990). Greigite has magnetic characteristics very similar to the Fe_7S_8 pyrrhotite: a Curie point of about 330–350 °C (Snowball 1971) and relatively high unblocking fields ($H_{ub\,max} > 100$ mT, for an IRM acquired in 1 T; Fig. 11 in Krs *et al.* 1990).

In the present sediments, the sulfides are most likely diagenetic minerals, witnesses of the original reducing environment. As such, the ferrimagnetic sulfide would be expected to carry a primary magnetization. In fact, it appears that a VRM has partly to totally replaced the original magnetization. A similar situation is observed by Kligfield & Channel (1981) who demonstrated that a VRM with maximum T_{ub} of 350 °C, carried by pyrrhotite grains close to the superparamagnetic threshold, totally remagnetized Mesozoic limestones. Krs *et al.* (1990) also pointed out that greigite was not a good recorder of palaeodirections; Salmon *et al.* (1988) reached the same conclusion for pyrrhotite.

The SE component is probably a CRM carried by magnetite cristallized sometimes later than the sulfide. In a palaeomagnetic study of marly limestones containing pyrrhotite and magnetite, Linssen (1988) suggested that no significant time delay separated the crystallization of the two magnetic phases, on the basis of the close proximity of the direction of the components carried by these two minerals. Here, we have no information concerning the direction of the primary remanence carried by the sulfide, except maybe in rare samples where the SE direction is reached as early of 300 °C, that is at a temperature lower than the highest possible T_{ub} for greigite. A possible explanation is that the Fe-sulfide grains with the highest T_{ub} have kept their original magnetization. If this is true, its direction being apparently identical to the SE direction carried by magnetite, we may conclude that this last component is also primary or almost so. But this observation relies on too few samples to be conclusive.

In the altered samples, the magnetic parameters strongly support that goethite, in the weakly altered ones, and haematite, in the most oxidized, carry the magnetization component. The presence of maghemite is also suggested in samples from Site C.

6 DISCUSSION AND GEODYNAMIC IMPLICATIONS

6.1 Arguments supporting a Givetian age for the MER pole

Among the seven magnetization components found in the Givetian rocks of this study, only the SE one is meaningful. Whether this magnetization is of primary origin cannot be unambiguously deduced even if the fold test is positive, because the tectonic tilting in the northern part of Ahaggar occurred at least 70 Myr after the rock formation. The difference in the pole position between 85 per cent and total tilt correction is only 1.5° (Table 3), but depending on whether the SE component is interpreted as a primary Givetian remanence or is considered to be syn-deformation and consequently of post-Moscovian age, brings about quite different continental drift implications.

Several arguments favour, or at least do not contradict, the hypothesis that the MER pole could be Givetian. First, there are indications that the magnetic carriers of the SE component (magnetite) might be diagenetic or nearly so. Then, the overall reversed polarity is consistent with the polarity time-scale given in Haq & Van Eysinga (1987) in which the middle Devonian time period is predominantly reversed. Also, this pole is significantly different from any other more recent ones, especially the Permian poles (Table 4 and Fig. 12), and lays in a plausible situation not far from several Carboniferous poles of Africa (Fig. 12); it places the sampling area in a subtropical palaeolatitude of −18.8°, consistent with the limestone formation. This VGP also implies that, during the middle Devonian, Africa was rotated about 90° clockwise relative to its present position, so that its NW coast was in intertropical latitudes, consistent with the middle Devonian reefs found in Morocco (Hollard 1967; Gendrot 1973).

6.2 Comparison with other Devonian poles from west Gondwana

A selection of the most reliable Devonian poles from Gondwana is presented in Fig. 12 and Table 4, together with poles from the Carboniferous to Triassic of western Gondwana. Here, we have omitted to report VGPs calculated from remagnetized directions to avoid circular reasoning, as these poles are generally dated by reference to an APWP which indeed is poorly established.

The Devonian poles from Africa are highly scattered and do not follow a chronological path; this is evidenced by the distance between the poles of middle Devonian age of the Gilif Hills Complexes, GC (Bachtadse & Briden 1991), the upper part of the Gneiguira Supergroup, GN2 (Kent, Dia & Sougy 1984) and the present study at Meredoua, if the MER pole is Givetian. Obviously, most of these poles, if not all, are erroneous.

The Gilif Hills pole is obtained on igneous rocks from ring complexes. The characteristic magnetization component C is probably an original TRM, as shown by the authors, but the palaeohorizontal is poorly constrained by flat-lying Mesozoic sediments which suggest that no Mesozoic or post-Mesozoic tilting occurred in the area. Also within-site and between-sites dispersion is large (Table 4) suggesting possible relative movements between the different places.

No fold test is available for the Gneiguira Formation, neither in the lower Devonian (GN1) nor in the middle Devonian part (GN2) of the section. The GN1 pole is obtained from the thrusted tectonic unit while the GN2 pole, which is determined from autochtonous and therefore more reliable formations, is based on only 12 samples.

The upper Devonian pole of Ben Zireg (Aïfa, Feinberg & Pozzi 1990) is also questionable; the characteristic magnetization considered to be primary (Famennian) is

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Table 4. Selected middle to upper Palaeozoic south poles for the Gondwana, in African coordinates (rotated, where appropriate, according to Smith & Hallam 1970)

Geological Formation	Symbol	Age	λ_p	ϕ_p	K	(A_{95}) dp, dm	$n(N)$	Reference
AFRICA								
Aïr Ring Complexes (Nigeria)	AI	435 My	-43.4	8.6	50	(6.2)	80 (12)	Hargraves et al. (1987)
Gneiguira redbeds (Mauritania)	GN1	low Dev	-38.3	43.8	118	(6.9)	32 (7)	Kent et al. (1984) Kent et al. (1984)
	GN2	mid Dev	-28.6	44.5	190	(9.0)	12 (3)	
Gilif Hills Ring Complexes (Sudan)	GC	377 ± 5	25.9	11.6	-	19.6 20.6	39 (11)	Bachtadse & Briden (1991)
Meredoua marly limestone (Algeria)	MER	mid Dev	-16.2	61.7	853	(4.2)	53 (3)	present study
Ben Zireg Griotte limestone (Algeria)	BZ	up Dev	-19.2	19.8	-	3.7 5.3	54 (3)	Aïfa et al. (1990)
Saïda + Béchar limestone (Algeria)	SB	mid-up Carb ?	-26.4	58.9	71	(9.1)	30 (5)	Abou Deeb & Tarling (1984)
Hassi Bachir + Aïn Ech Chebbi redbeds (Algeria)	HA	up Carb	-25.4	54.8	149	(3.5)	43 (11)	Daly & Irving (1983)
Illizi sediments (Algeria)	IL	up Carb	-28.7	55.8	235	(2.9)	61 (10)	Henry et al. (1992b)
Dwyka varves (Tanzania, Zambia)	DW	up Carb ?	-26.9	30.3	73	(10.8)	25 (4)	Mc Elhinny & Opdyke (1968)
Tiguentourine redbeds (Algeria)	TI	Carb/Perm	-35.3	60.3	175	(3.5)	62 (10)	Henry et al. (1992a)
Abadla redbeds (Algeria)	AB	low Perm	-29.2	59.6	-	(5)	48 (13)	Morel et al. (1981)
Chougrane redbeds (Morocco)	CH	low Perm	-32.2	64.1	-	2.3 4.7	45(1)	Daly & Pozzi (1976)
Taztot Trachyandesite (Morocco)	TT	low Perm	-38.7	56.8	-	2.3 4.6	12 (1)	Daly & Pozzi (1976)
Kenifra, Jerada, Beni Snassen, Tazzeka (Moroc.)	KJST	mid Trias	-69.8	71.6	44	(11.7)	54 (5)	Daly & Pozzi (1976)
Aït Aadel dolerites (Morocco)	TM1	mid Trias	-72.0	74.5		(7.5)	63 (15)	Hailwood (1975)

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Table 4. (*Continued.*)

Geological Formation	Symbol	Age	λ_p	ϕ_p	K	(A_{95}) dp, dm	$n(N)$	Reference
Tholeiites (Morocco)	TM2	up Trias	-72.0	38.0	17	(6)	240 (34)	Bertrand & Westphal (1977)
Argana redbeds (Morocco)	AR	up Trias	-50.6	71.4	-	(8.0)	22 (22)	Martin et al. (1978)
Rhyolite + subvolcanics (Egypt)	RS	216 ± 5	-64.1	50.7	-	2.6 5.1	50 (10)	Saradeth et al. (1989)
Sakamena Group (Madagascar)	SG	up Trias	-58.0	86.0	-	(6.3)	31 (7)	Mc Elhinny et al. (1976)
SOUTH AMERICA								
La Collina Paganzo	CP1	low Perm	-46.3	67.6	-	4.0	35 (1)	Thompson (1972)
La Collina Paganzo (Argentina)	CP2	266 ± 7	-59.7	68.1	-	3	129 (1)	Valencio et al. (1977)
AUSTRALIA								
Snowy River Volcanics (Lachlan Fold Belt)	SR	low Dev	-48.2	13.4	13.3	(13.7)	47 (10)	Schmidt et al. (1987)
Comerong Volcanics (Lachlan Fold Belt)	CV	up-mid Dev	-25.8	5.8	41.5	7.2	52 (11)	Schmidt et al. (1986)
Hervey Group (Lachlan Fold Belt)	HG	up Dev	2.5	13.9	17.2	(13.8)	56 (8)	Li et al. (1988)
Canning Basin limestone (NW Australia)	CB	up Dev	7.8	22.5	86.5	(6.5)	89 (7)	Hurley & Van der Voo (1987)
Mount Eclipse sandstone (Central Australia)	MES	mid Carb	-17.6	78.7	-	19.2 19.7	? (16)	Li et al. (1989)
Main Glacial Stage (New South Wales)	MG	up Carb	-39.4	67.2	16.2	(9.8)	71 (15)	Irving (1966)

The symbols are those used in Table 1; n and N represent respectively the number of samples and the number of sites.

much better grouped after than before tilt correction and therefore the fold test is claimed to be positive by the authors. In detail, however, the evolution of the three site mean directions from before to after unfolding (Fig. 8 of Aïfa *et al.* 1990) reveals two inconsistencies which might reflect a more complicated tectonic story than presently assumed: at Site BE which is at the hinge of the fold, even after a total structural correction, the mean direction does not manage to reach the tilt-corrected direction of the two other sites (BJ and BZ), as if the correction applied was not sufficient. Conversely, at sites BJ and BZ where the dips are roughly opposite, the mean directions cross each other beyond 85 per cent of unfolding, as if the dips were overcorrected. Here, contrary to what we observed in our study, the bivariate statistics show a clear evolution of K, α_{95} and the shape of the ellipse associated to the mean direction upon progressive unfolding of Sites BJ and BZ; very elongated when no tilt correction is applied ($\alpha(y) = 3.5$; $\alpha(x) = 12.9$), the ellipse becomes close to circular for 85 per cent of unfolding ($\alpha(y) = 3.2$; $\alpha(x) = 3.3$), then more elliptic again, for a total fold correction ($\alpha(y) = 3.2$; $\alpha(x) = 3.9$). The other component (not reported here), interpreted by Aïfa *et al.* (1990) as a Tournaisian (early Carboniferous) remagnetization, gives a

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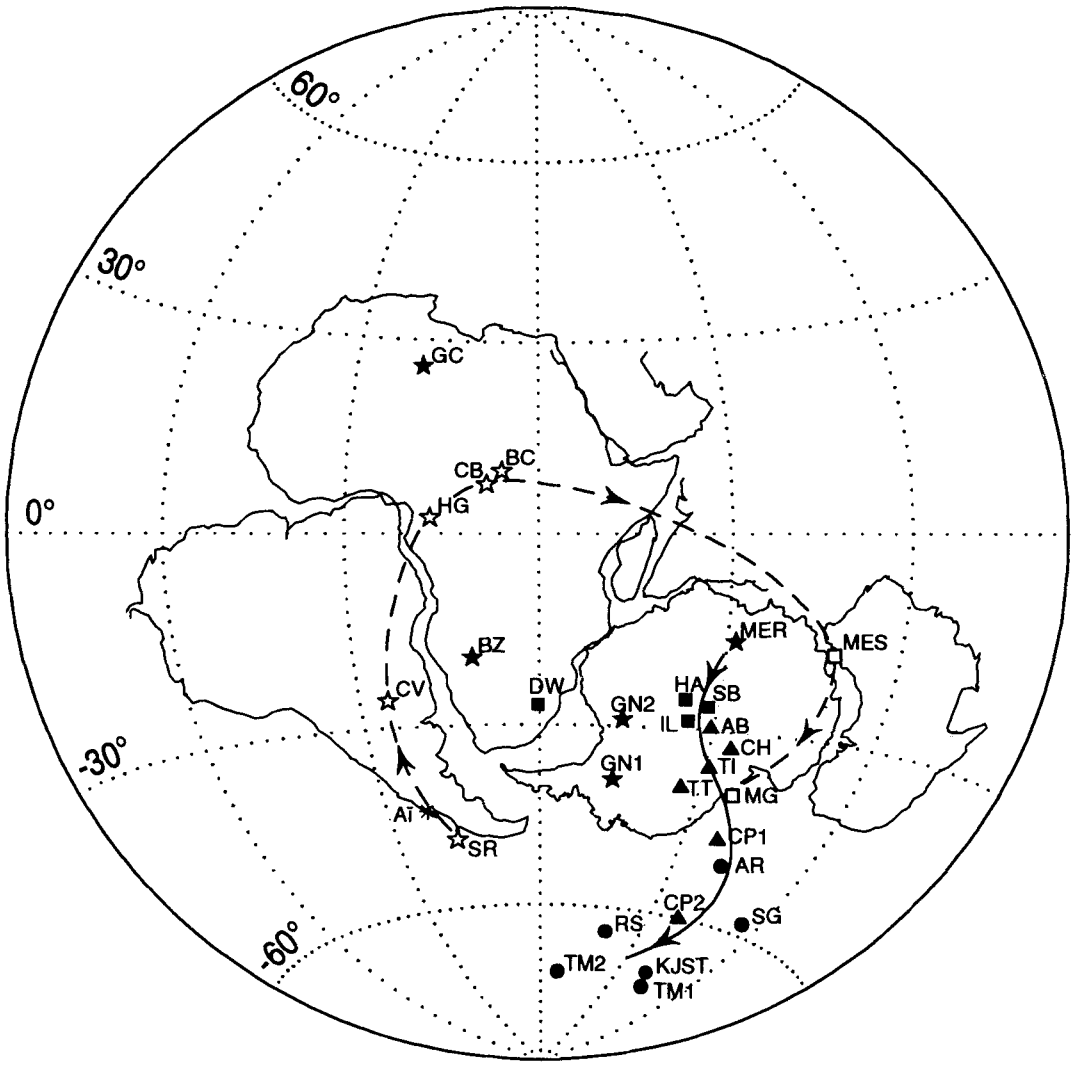


Figure 12. Palaeomagnetic South poles for East and West Gondwana in African coordinates according to the Smith & Hallam (1970) reconstruction. Open symbols refer to Australia: stars and squares represent respectively the Devonian and the Carboniferous poles. Solid symbols refer to western Gondwana: stars, Devonian; squares, Carboniferous; triangles, Permian; circles, Triassic. The dashed and full lines represent respectively the APWP for eastern and western Gondwana. See Table 3 for an explanation of the symbols.

best grouping of the directions after a differential and almost complete unfolding. But their Famennian and Tournaisian directions are not significantly different at the statistical level, either in geographic or in stratigraphic coordinates, and even after a partial tilt correction of the Tournaissian data. Concerning the tectonic history and setting of the area, the Devonian of Ben Zireg outcrops in a pinched anticline of Palaeozoic rocks pervasively affected by axial plane cleavage associated to minor folds observed at all scales (Ball *et al.* 1975). This implies a severe internal deformation of the rock and consequently a probable distorsion of the original remanent magnetization if it has been preserved. A first tectonic activity of upper Tournaisian or lower Visean age is suggested from a sedimentary unconformity between upper Visean and Tournaisian strata (Pareyn, Lemosquet & Weyant 1975); but the major tectonic phase would be post upper Visean and responsible for the flow cleavage which affects the series up to the upper Visean rocks (Fabre 1976); post-Namurian shallow-dipping thrusts are also documented in the area (Ball *et al.* 1975); thus, the first important tectonic phase occurred definitely later than the Tournaisian, contrary to the statement of Aïfa *et al.* (1990). Consequently, even if their fold test could be considered to be positive, it does not constrain the age of the characteristic magnetization of the Ben Zireg griotte limestones to be Fammenian.

6.3 Discussion about the Carboniferous poles of West Gondwana

The Carboniferous Dwyka pole, DW (McElhinny & Opdyke 1968), whose precise Carboniferous age is still debated, is remote from the three other tightly grouped ones (poles SB, HA and IL in Fig. 12); we have recalculated this pole, from McElhinny and Opdyke's data, using the results of their sites A_1, A_2, B_1 and B_2, but excluding Site C because the thermal treatment is incomplete and the present field component probably not totally removed. In general,

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this study suffers from an insufficient demagnetization procedure (the technique of pilot samples was used) and from a lack of graphical representation for the demagnetizations. The 25 samples used for the calculation provide a beautiful fold test which unfortunately is not very constraining because the tilting did not occur before the Tertiary.

The SB pole calculated from the data of Abou Deeb & Tarling (1984) on the five sites selected by the authors for their best grouped directions is also doubtful because three out of five sites are not precisely dated: they are assigned a Carboniferous age with no more precision; the two others are Namurian (lower middle Carboniferous) and Westphalian (late Carboniferous) respectively. Also, nothing is said about the tectonic context of the sedimentary formation nor about the coordinate system in which the directions are given.

The two last Carboniferous poles IL and HA appear to be more reliable, although no fold test is available for the Illizi Formation (Henry *et al.* 1992b). The average pole of Hassi Bachir and Aïn Ech Chebbi (Daly & Irving 1983) is better defined after than before tilt correction, suggesting a positive fold test, where the folding is partly late Carboniferous or early Permian and partly upper Jurassic in age. But here again, a progressive tilt correction gives an optimum unfolding around 70 per cent when the 11 sites are considered separately, and at 75 per cent when the sites with identical dip are grouped together; as in our study, the K and α_{95} curves display respectively a broad maximum and a wide minimum upon stepwise unfolding, so that the same question as in the present study can be asked as to whether this magnetization is ante- or syn-deformation.

6.4 APWP for West and East Gondwana in the upper Palaeozoic

Selected Permian to Triasic poles from western Gondwana are also presented in Fig. 12. If we exclude the Dwyka pole (DW), an APWP for West Gondwana can be traced from Moscovian (late middle Carboniferous) to Triassic. This curve can be extended back in time to the middle Devonian if a Givetian age is assigned to the MER pole. Then, the poles follow each other approximately in a chronological order, describing a smooth curve which expresses mainly a northward latitudinal drift of Gondwana together with an anti-clockwise rotation of about 45° to 50°. This curve implies that between the upper Givetian and the Moscovian the minimum drift of West Gondwana was about 1.5 cm yr^{-1}, comparable to the actual continental drift rates.

Because of the lack of reliable middle Palaeozoic poles from Antarctica and India, the APWP of eastern Gondwana is based only on Devonian and Carboniferous poles from Australia (Fig. 12). Rotated relative to Africa, according to the Smith & Hallam's (1970) reconstruction, these poles are chronologically distributed along a regular curve, but they are always significantly different from African poles of similar ages. The Australian path implies an anti-clockwise rotation of about 50° of East Gondwana during the Devonian, the Euler pole being located slightly north from the present northern coast of Australia, then a fast southward latitudinal drift of some 60° from the upper Devonian to the middle Carboniferous (about 20 cm yr^{-1} of minimum drift rate).

Among the Devonian poles of Australia selected in Fig. 12 and Table 4, only two of them, of upper Devonian age, come from a cratonic area: the Canning Basin, CB (Hurley & Van der Voo 1987) and the Brewer Conglomerate, BC (Chen *et al.* 1993); the others, slightly older (CV, SR and HG), originate from the south-eastern orogenic margin of New South Wales (Commerong Volcanics, Schmidt *et al.* 1986) and the Lachlan fold belt (Snowy River volcanics, Schmidt *et al.* 1987; Hervey Group, Li, *et al.* Schmidt & Embleton 1988) whose docking to the Australian craton is not demonstrated before the mid-Carboniferous (Powell, Li & Thrupp 1990). Consequently, the poles SR, CV and HG might not be representative of the stable part of Australia and thus of eastern Gondwana.

Let us now assume with Chen *et al.* (1993) that the BC pole corresponds to a primary magnetization of late Devonian age, although the optimum unfolding is 91 per cent according to the K and α_{95} values, as previously mentioned; with the same logic, let us accept a Givetian age for the MER pole. The cratonic Australian poles (CB + BC) and MES imply a rapid drift of East Gondwana between the upper Devonian and the middle Carboniferous, which is at variance with the low drift rate of West Gondwana infered from the MER pole and the group of African Moscovian poles for about the same period of time.

If we examine the Devono-Carboniferous data of Australia, there seems to be no doubt about the reliability of the Frasnian to Famennian age of the Canning Basin pole CB, although its situation might not be very accurate, due to uncertainties about the original syn-sedimentary dip at some sites (Hurley & Van der Voo 1987). The Brewer Conglomerate has a similar age or is slightly younger than the Canning Basin Formation, and its pole lies very close to the CB one (Fig. 12). Therefore, it probably corresponds to a primary remanence of late Devonian age.

The cratonic Mount Eclipse Sandstones pole, MES (Li *et al.* 1989), is considered by the authors to be syn-deformational, but the best grouping of the directions, which the authors consider to be the most reliable, is always obtained after a differential unfolding in two steps of the northern and southern limbs and after a global tilt correction of 75 per cent for seven sites and 35 per cent for the nine others. But still, the oval of confidence is quite large (see Table 4). The fundamental question is whether this complicated unfolding process reflects the real geometry of the fold at the time of the remanence acquisition, although the same question may also be asked when a more simple proportional unfolding is applied.

We have recalculated the upper Carboniferous pole, MG, of Irving (1966), using only the 15 sites of the Hunter Valley, Currabubulla and Rocky Creek Formations in which a better grouping of the directions is obtained after than before tilt correction. We note that this pole is not very remote from the African poles of about similar age (HA, IL, SB), but it is closer to the lower Permian poles of West Gondwana (Fig. 12).

Let us now admit that the MER pole is not Givetian and that the West Gondwanian poles of Fig. 12 are reliable for the mid-Carboniferous to Triassic interval of time. As already underlined, the SE magnetization component may

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have been acquired any time between the upper Givetian and at least the Moscovian, even if the fold test is positive. Considering the location of the MER pole relative to the West Gondwanian APWP of Fig. 12 and given that this pole is significantly different from the upper middle Carboniferous African poles HA and IL (Table 4), the age of the MER pole would rather be older than middle Carboniferous. Then, the SE remanence would be a CRM generated through some kind of weathering process, as no structural deformation nor thermal event able to create a CRM between the Givetian and the Moscovian is known in this area.

The last possibility is that the fold test is negative and the SE remanence is syn-deformation; consequently, the MER pole is necessarily post-Moscovian, that is probably upper Carboniferous or Permian, but then, the younger this pole, the greater the distance with West Gondwanian poles of similar age (Fig. 12). So, if an age immediately posterior to Moscovian is still conceivable for this pole, a Permian age can be ruled out, unless all the Carboniferous and lower Permian poles of the West Gondwania APWP of Fig. 12 are remagnetized. An argument supporting this last hypothesis could perhaps be found in the HA pole of Daly & Irving (1983), if the magnetization is in fact syn-folding, as previously envisaged. The deformation which is late Carboniferous to early Permian at Hassi Bachir and Aïn Ech Chebbi is likely to be contemporaneous about 200 km south at Meredoua (present study). If the characteristic magnetization is syn-folding in both studies, then the HA and MER poles would be rejuvenated around late Carboniferous or early Permian. But if they had a similar age, these poles should not be statistically different as they are. According to this logic, the other Carboniferous poles SB and IL would have been also remagnetized at the same time. Unfortunately, this hypothesis cannot be tested on these poles because no fold test is available. This interpretation is not contradicted by the lower Permian pole AB of Abadla (Morel *et al.* 1981) which is not statistically different from the SB and IL poles (Fig. 12); however, most of the lower Permian poles are significantly distinct from the previous group (HA, IL, SB and AB): TI and TT for Africa, but more particularly CP1 and CP2 for South America. In this interpretation, the upper Carboniferous pole MG of Irving (1966) would be also remagnetized in the Permian times.

7 CONCLUSION

This study has emphasized the need to look carefully to the basic data and to perform stepwise unfolding, when possible, to relate more precisely the age of both the magnetization and the deformation. But it has also underlined the difficulty in interpreting the test in some limit cases where the optimum unfolding is almost total and the curves of K and α_{95} do not display a well-defined maximum and minimum concentration respectively. It points to the need of performing fold test preferably on steeply inclined strata. As a consequence this study led us to re-examine previous data where the fold test was claimed to be positive on the sole basis of a best grouping of the directions after than before tilt correction.

The interpretation of the present data is not obvious. The optimum tilt correction is reached for 85 per cent of unfolding, but the fold test may probably be interpreted as being positive, in view of the spread shape of the K and α_{95} curves and considering the likely slight original syn-sedimentary dips at the different sites.

Assuming that the MER pole is Givetian in age, let us focus on the middle Devonian time period and look at the geodynamical implications of the corresponding poles. Assuming that Laurussia was situated in the latitudinal position given in Van der Voo (1988) and Kent & Van der Voo (1990), the palaeocontinental reconstruction based, for the Gondwana, on the non-cratonic Australian pole, CV, of Schmidt *et al.* (1986), implies the existence of a large pre-Variscan oceanic domain (the Rheic Ocean) between the northern continents of Laurussia and the southern ones of Gondwana. This ocean is even wider (about 7000 km) if the cratonic African pole, GC, of Bachtadse & Briden (1991) is used. In contrast, the African poles GN1 and GN2 of Kent *et al.* (1984) and MER of the present study imply that the Rheic Ocean was already nearly closed in middle Devonian times. This last hypothesis is consistent with the increasing benthic faunal affinities observed by Morzadec, Paris & Racheboeuf (1981), Morzadec *et al.* (1988) between Laurussia and Gondwana in the course of the Devonian; it also supports the palaeogeographic reconstructions proposed by Robardet, Paris & Racheboeuf (1990) and by Paris & Robardet (1990), based on synthetic studies of sedimentological facies and faunal assemblages. But the hypothesis according to which the MER pole would be Givetian strengthens the discrepancy between the East and West Gondwanian APWP segments for the middle Devonian to the middle Carboniferous interval of time.

If the fold test cannot be accepted as positive and if the magnetization is syn-tilting, as the structural deformation is post-Moscovian in the studied area, we are led to rejuvenate at least all the middle Carboniferous poles of West Gondwana in order to get a coherent evolution pattern of the APWP.

We rather believe that most of the West Gondwanian Carboniferous and Permian poles of Fig. 12 are reliable and that our fold test is positive, implying that the SE magnetization component is ante late Moscovian. But without conclusive argument in favour of a primary origin of this magnetization, the MER pole may have any age between the upper middle Devonian (Givetian) and the upper middle Carboniferous (Moscovian).

If the MER pole is younger than Givetian, e.g. upper Devonian or lower Carboniferous, the palaeocontinental reconstruction is not changed fundamentally relative to the previous one, first because the MER pole is not very remote from the African Moscovian poles, then because the northward drift of Laurentia from the middle Devonian to the lower middle Carboniferous is very limited according to Scotese & Golonka (1992). Then, the palaeocontinental assembly is consistent with the Pangea B reconstruction of Irving (1977), as already stated by Daly & Irving (1983) and Henry *et al.* (1992b) and with the formation of the Variscan orogeny.

Without excluding the possibility that the MER pole is Givetian in age, another tentative interpretation could be proposed: this pole is not very distant from the extrapolated intersection of the APWPs of East and West Gondwana in

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Fig. 12. It is located in the vicinity of the Australian curve, between the upper Devonian BC pole and the middle Carboniferous MES pole; consequently, it could be estimated to be slightly older than middle Carboniferous. A lower Carboniferous age or nearly so would be the most consistent relative to both APWP curves; but is this interpretation meaningful when such a large distance exists between the middle Carboniferous Australian pole (MES) and the African ones (HA, SB, IL) of comparable ages? In other words, are these two APWP segments reliable? May they be used to draw a common curve for the whole Gondwana where the West Gondwanian path would chronologically follow the East Gondwanian one, or should East and West Gondwana be considered separately?

Clearly, many more poles are still needed for the Devono-Carboniferous time period in order to test the reliability of Australian poles on the one hand and African ones on the other, and simultaneously check the commonly accepted hypothesis of the existence of an East and a West Gondwana behaving as a unique rigid block throughout the whole Palaeozoic era.

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